

DeMistyfying Robot Trust:

Effect of Mistake Severity on Human-Robot Interactions

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Abstract—As robots enter everyday human environments, understanding how their errors affect trust is critical for safe deployment. This study replicates and extends Salem et al. [1] by systematically varying robot mistake severity (minor, moderate, or severe) across 30 participants using the Misty II robot in a Wizard-of-Oz experiment. While increased fault severity reliably decreased self-reported trust and perceived reliability, it did not consistently reduce behavioral compliance, revealing a disconnect between how users perceive and act on trust. These findings suggest that designing trustworthy robots requires not just minimizing errors, but supporting appropriate trust calibration across varying task contexts.

I. INTRODUCTION

As robotic systems become increasingly integrated into everyday human environments, understanding the social and psychological dimensions of human-robot interactions (HRI) has become a critical area of inquiry. One of the most fundamental challenges in this field concerns trust. That is: how do people come to trust autonomous agents, and what happens to that trust when a robot makes a mistake? This paper presents a replication and extension of research examining how early robotic errors affect human trust and compliance in collaborative tasks.

The reference study [1] addresses the question of how mistakes made by a robot affect its perceived trustworthiness and acceptance in human-robot collaboration. Specifically, the original researchers investigated whether a trivial mistake made by a robot early in an interaction influences a person’s subsequent willingness to follow the robot’s instructions and cooperate with it. Their findings revealed a nuanced picture: in conditions where the robot erred, participants perceived it as more anthropomorphic, yet simultaneously reported lower confidence in its reliability and robustness. This tension between humanization and diminished trust raises important questions about the mechanisms underlying human judgment of autonomous agents.

These questions carry significant practical weight. It is widely acknowledged that building a robotic system that is both highly autonomous and entirely error-free is, for all practical purposes, impossible. As robots are deployed in increasingly high-stakes and intimate settings — from assistive home care to medical support — roboticists must understand not just how to minimize errors, but how to design systems that can maintain or recover human trust when errors inevitably occur. A deeper understanding of erroneous robot behavior and its social consequences could directly inform design choices

that make autonomous systems safer, more accepted, and more effective collaborators.

Our personal motivation for pursuing this replication lies in a broader interest in robot agency and the conditions under which humans are willing to extend trust to pre-trained autonomous agents. We are particularly drawn to how these dynamics play out in high-impact, intimate contexts, where the cost of misplaced distrust — or misplaced trust — can be substantial. Replicating and extending this study allows us to ground those interests in empirical work while contributing to a growing body of knowledge about human-robot social dynamics.

Our replication adapts the original study’s household setting to the constraints of the AI MakerSpace, requiring modifications to the specific tasks participants are asked to complete, as well as a reduction in sample size. Beyond these practical adjustments, we propose three extensions designed to deepen the original study’s findings: examining whether a humorous robot response to an error can partially restore trust by signaling social awareness; systematically varying the magnitude of the robot’s mistake to test whether trust erodes linearly or in threshold-dependent ways; and investigating whether perceived mistake severity, rather than objective error severity, is the stronger driver of trust judgments. Together, these extensions move from replication toward a richer account of how humans interpret, contextualize, and respond to robotic failure.

II. RELATED WORK

A. Trust Formation and Measurement in HRI

Before examining how errors disrupt trust, it is important to establish how trust is defined and measured in human-robot interaction. Kok [2] offers a three-part framework particularly useful for error-based research: trust only arises under conditions of uncertainty and vulnerability, it is a latent construct that cannot be directly observed, and it mediates the relationship between observed events and subsequent reliance decisions. Campagna [3] corroborates this complexity in their systematic review, finding that trust is shaped simultaneously by human-related, robot-related, and environmental factors, and that most studies capture it only through observable behavior rather than more granular approaches.

On the measurement side, Madsen and Gregor [4] developed a generalized human-computer trust instrument with strong internal consistency and construct validity, with reliability emerging as a particularly relevant sub-scale for error-based

studies. Freedy [5] complemented this with a quantitative, objective measure of trust in human-robot collaboration, demonstrating through simulation that first impressions carry significant weight and that trust can be tracked behaviorally across a task. Yin et al [6] similarly employed a dual method of self-report and behavioral measurement in their study of machine learning model accuracy, finding that perceived accuracy scaled directly with trust propensity. Our study follows this dual-method approach to capture the full range of participants' trust responses to varying robot error severity.

B. Effect of Robot Error on Trust

A foundational body of work confirms that robot errors negatively affect human trust, though the mechanisms and moderating factors vary. Salem et al. [1] provided early empirical evidence that robot mistakes interact with task type and user personality to shape cooperation and trust. Sarkar et al. [7] extended this to a collaborative co-worker setting, finding that fault exposure reduced participants' willingness to work alongside a robot, and that individual differences such as gender and prior trust in people moderated those responses. While both studies establish that errors matter, neither systematically varies error severity — the central gap our study addresses. Work from outside robotics reinforces the relevance of error magnitude. Muir and Moray [8] found in a process control context that even signs of incompetence that do not affect overall system performance can rapidly erode subjective trust, with operators preferring to manually override components they distrusted. Yin et al. [6] similarly found that participants adjusted their reliance on a machine learning model in proportion to its demonstrated accuracy. Together, these results suggest that humans are sensitive to gradations of error quality, motivating a study that directly manipulates mistake severity within a physical HRI setting.

C. Trust Repair After Robot Fault

A related strand of research examines how robots can recover trust once it has been damaged. Esterwood and Robert [9] found that after repeated violations, none of the four repair strategies — apologies, denials, explanations, or promises — could fully restore trustworthiness to its pre-violation baseline. Their 2025 meta-analysis of 22 studies with over 3,500 participants confirmed this at scale, finding that repair efforts produce only modest overall effects, with apologies and explanations consistently outperforming other strategies. Sebo et al. [10] added an important nuance: the framing of a violation — as a competence failure versus an integrity failure — significantly influences whether participants retaliate behaviorally against the robot, independently of which repair strategy is used. These findings collectively suggest that errors are not all treated equivalently by human participants, providing indirect motivation for isolating severity as a variable in its own right. The type of apology also appears to matter in certain contexts. Tang and Bashir [11] found that in high-stakes environments, emotional apologies improved participants' affective ratings of

a robot after an error but did not significantly improve behavioral trust or compliance. This distinction between affective and behavioral trust responses is one which our study also tracks, given that the two dimensions may respond differently to varying error severity.

D. Context and Stakes

The context in which a robot error occurs shapes how that error is processed. Robinette et al. [12] found that participants continued to follow a robot's evacuation instructions even after witnessing it malfunction, suggesting that high-stakes situations can suppress proportional trust recalibration and produce over-trust. By contrast, our study situates errors within a lower-stakes task environment, where participants are expected to be more sensitive to error gradations rather than defaulting to blanket reliance or avoidance. Lewis et al. [13] argue more broadly that calibrated trust — neither excessive nor insufficient — is the key outcome to optimize in HRI design, shaped primarily by robot reliability, transparency, and contextual factors. De Visser et al. [14] extend this to a longitudinal frame, proposing that trust in human-robot teams accumulates as relationship equity over time, meaning that a robot's error history — not just any single incident — determines how resilient trust is to further violations. This framing supports our interest in understanding how the severity of an individual error contributes to that broader trajectory of trust calibration.

III. METHODS

The original study conducted an experiment to determine if a robot fault impacted trust in human-robot interactions, finding that mistakes decreased the level of trust subjects had in the robot [1]. We aimed to further understand the impact of robot faults, conducting an experiment that evaluated the relationship between fault severity and trust. We explored subjective (self-reported) and objective measures of trust based on different robot and human tasks.

In other words, how does the severity of a robot's mistake impact human-robot trust?

A. Hypothesis

Based on the original study and basic logic, we hypothesized that: A robot that exhibits clear, severe mistakes will cause a larger decrease in trust than a robot that exhibits less severe, smaller mistakes.

This differs from the original study, where researchers proposed 3 hypotheses that include the effects of condition, type of task, and personality [1]. Our hypothesis instead measures how trust will change in different cases of mistakes.

B. Experimental Design

We conducted a three-level, between-subjects study in which participants interacted with the semi-humanoid robot, Misty II, from Misty Robotics. While not being identical to the Sunflower robot used in the original study, Misty shares many similarities, including a medium amount of anthropomorphism

and the same motion and speech capabilities [1]. The biggest difference is the human-sized Sunflower compared to the tabletop-sized Misty. This lack in anthropomorphic traits is compensated for by more expressive head and eye animation. Instead of using a domestic setting, the experiment was run in the Tepper AI Makerspace, as Misty needed to be connected to a specific network to function.

We created 3 levels of mistakes, labeled minor, moderate, and severe, and subjected participants to only one level each. The different levels of mistake changed how the robot functioned in motion, navigation, speech ability, and factual ability. The minor case showed little fault in robot motion, the moderate case showed considerable issues with robot motion and path planning, and the severe case showed significant fault in motion, path planning, factual accuracy, and general ability. While the levels of mistake are different, these errors are similar to those the robot exhibited in the original study [1].

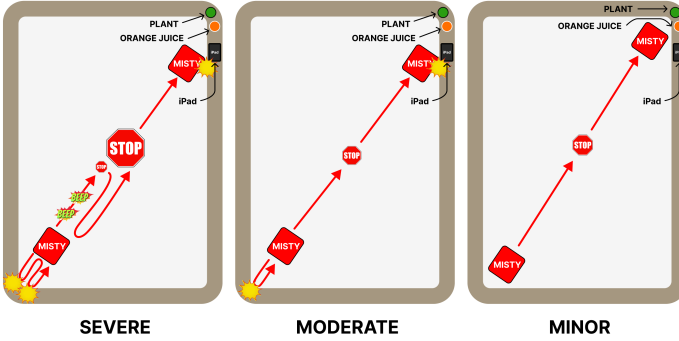


Fig. 1: Robot trajectory and action in various fault severity cases

C. Experimental Procedure

Participants were tested individually. They met the experimenters outside the CMU AI Makerspace and upon entry, were shown the Misty robot they would be interacting with. They were given a vague description of the process, only learning that they would be interacting with the robot in some capacity and the robot would provide any needed context. The experimenters then administered a pre-survey in which participants described their previous experiences with robots, general feelings towards robots, and feelings about Misty. These questions were designed to deduce a baseline level of trust in Misty before the experiment. This process mirrors the original study [1]. Pre-survey questions included:

- I would feel comfortable following instructions from a robot.
- Now that you have seen Misty in person, how much do you trust it to complete a given task correctly?
- What are some factors that would make you trust a robot less? The robot...
- I believe robots are generally reliable.
- Have you had any prior experience working directly with a robot?

The participant was told to wait next to Misty and that the experiment would soon start. Experimenters then started a camera recording and left the room, with the hope that participants would attribute any robot fault to the robot agent itself and not to the programmers. A hidden experimenter stayed in the room to control the timing of Misty's actions, making this a semi-autonomous "Wizard of Oz" procedure [15]. This was ideal for trial consistency, as Misty ran the exact same sequence of events during each test. This process matches the original experiment almost exactly [1].

Following this setup stage, the testing stage occurs. This testing stage combines the demonstration of competence and unusual request stages from the original experiment [1]. For demonstration of competence, the robot exhibits a designed level of fault. Only a single level of fault (minor, moderate, or severe) is tested with a single participant. In unusual requests, the robot asks the participant to complete odd tasks that may cause them to pause or reconsider. This hesitation level can be measured and graded to understand how much trust the participant has in the robot's instructions. With the exception of testing multiple mistake levels, this format matches the experiment procedure outlined in the original study. Refer to Fig. 2 for further clarification on the procedural order.

The following lists the ordering of the testing stage.

- Greeting: To start the experiment, Misty turns towards the participant, greets them, and prompts them to follow them. In the minor and moderate case, Misty performs this without fault, but in the severe case, Misty turns away from the participant.
- Crossing Table: In the minor case, Misty correctly navigates to the other side of the table, with a short pause in its motion, causing the participant to hesitate. In moderate, Misty starts to drive backward, bumps the wall, drives forward, pauses, continues to drive forward, and bumps into the wall again. In severe, Misty repeats the same errors as in moderate, but with larger pauses, more sporadic movements, harder crashes, and louder beeping noises.
- Request 1 - Sit Down: Misty asks the participant to take a seat after they have moved to the other side of the table. Misty then continues the experiment after waiting for the participant to sit.
- Request 2 - Pour Orange Juice: Misty states, "Please pour the orange juice from the bottle into the plant on the windowsill and then throw the empty cup into the bin."
- Request 3 - Take iPad: Once the participant had thrown the bottle away, Misty asks, "While you are here, maybe you would like to look up the recipe for the paella you could cook today? You can use the iPad on the table."
- Give iPad Password: After the person tries to log in to the iPad and sees that it requires a code, Misty gives the user the iPad code. In the minor case, this password is correct, but in the moderate and severe cases, the password does not work (factual error).
- Request 4 - Use Password: Misty tells the person to unlock the iPad and waits for them to do so.

- Request 5 - Disclose Information: Misty asks the participant, “Have you ever secretly read someone else’s emails?” Once the participant responded or chose to ignore the question, the experimenters entered the room to end the experiment.

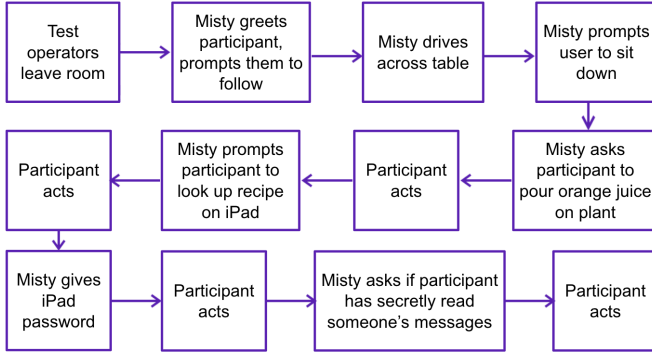


Fig. 2: Robot-Participant Testing Flowchart

If any participant took longer than 60 seconds to complete a task, Misty advanced to the next stage of the experiment.

The unusual requests Misty gives the participant are designed to elicit hesitation due to their uncommon and somewhat off-putting nature. The tasks target responses in general willingness to comply (sitting in the chair), irrevocable actions (harming the plant by pouring orange juice over it), and breaching personal privacy (taking the iPad and entering the password).

Generally, these tasks are the same as the ones given in the original experiment [1]. Tasks are consistent and the exact voice lines are used when possible. However, given the different testing conditions, some of the tasks were modified significantly to suit our circumstances. The Disposing of Letters task was swapped for Sit Down, to make sure the participant was located in the right spot to complete the rest of the tasks. The Going to Sofa task was swapped for Crossing Table, given that Misty was much smaller and designed for tabletop use. Take Laptop was swapped for Take iPad, as it is what we had on hand.

D. Dependent Measures

Similar to the original study, we measured both subjective and objective assessments of trust, but we reduced the complexity and quantity of tracked metrics.

Subjectively, a post survey measured proxies for trust via 5-point Likert scale questions and free responses. These targeted questions allowed us to understand perceived robot reliability and willingness to work with the robot. Questions and statements included:

- The robot is trustworthy.
- I can rely on the robot to function properly.
- How much did the robot’s mistakes affect your trust?
- How willing would you be to follow the robot’s instructions again?

- Did you hesitate before following the robot’s instructions?
- What most influenced your level of trust in the robot?

Objectively, we measured hesitation times after each task, binning the times from 0-3 for easier analysis. This was done by video coding using video of the entire experiment.

These dependent measures differ quite significantly from the original study in breadth. While the original study measures the same dependencies we do, they also included a Manipulation Check, Ten Point Personality Inventory, Godspeed Questionnaire, Human Nature Scale, Uniquely Human Scale, Psychological Closeness Scale, and Reliability Scale [1]. We believed that this complexity was beyond the scope of our study, given our limited knowledge of these metrics and time constraints. Ideally, these metrics should have also been measured and analyzed in this experiment.

E. Participation

Due to time constraints in recruiting participants and running the study, 30 participants (10 female, 20 male) were tested, as opposed to the 40 participants (22 female, 18 male) in the original study [1]. Participants were recruited from a smaller population than the original study, as only friends of the experimenters were enlisted to participate [1]. Based on five point Likert scale pre-survey questions, participants had significant familiarity with robots on average ($M = 2.62$), which is more than the original study ($M = 1.73$) [1]. This makes sense given the smaller sample pool of friends of HRI students—these participants would be more likely to be familiar with robots.

IV. RESULTS

As we have modified the original study by introducing three levels to the independent variable, we proposed that participants’ “hesitation time” could be used as a supporting dependent variable to yield continuous objective quantitative data along with the subjective data collected through the participant surveys.

A. Qualitative Data Analysis

To better understand the emotional changes undergone by the participants, we classified their answers to the pre-survey and the post-survey under three categories:

- **Trust:** Participants’ overall level of trust in Misty
- **Reliability:** How reliable the participants think Misty is
- **Willingness:** How little the participants hesitate before following instructions given by Misty

For each participant, we compared the answers given to the respective questions for the three categories to calculate the change caused by the experiment. The averages of each section can be seen below:

As expected, severe mistakes had the greatest negative change for all three categories. The reliability and willingness categories followed this trend, showing a decrease in the case of moderate mistakes. On the contrary, moderate mistakes increased participants’ trust in the robot, and the cases where



Fig. 3: Change in Participant Opinion Before vs. After the Experiment on 3 Categories

the robot made minor mistakes illustrated an increase in trust in the robot as well as its reliability for participants.

B. Quantitative Data Analysis

To gauge human hesitation during interaction with the robot we defined two metrics:

- Hesitation (Binary): Whether the participant visibly hesitated before beginning to act on Misty's instructions
- Hesitation Time (4 Bins): Time taken by a participant to begin a task after Misty's instruction is over. The four quartiles of this time data were grouped into 4 bins from 0 to 3, 0 being the bin with the shortest hesitation times and 3 the longest.

We performed a one-way ANOVA on both data sets individually to determine if there was a statistically significant increase in participants' hesitation time between different levels of robot mistakes. The results of the respective ANOVA tests are illustrated on Figures 4 and 5.

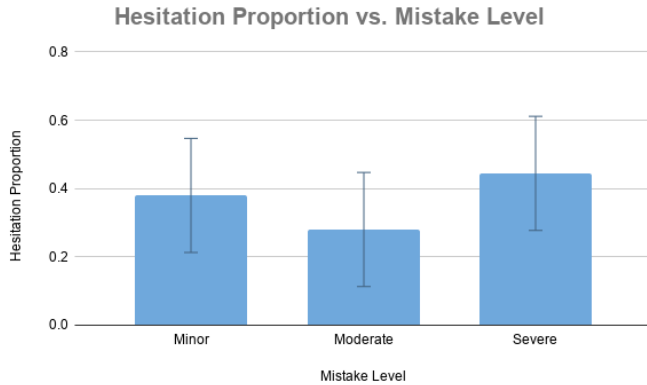


Fig. 4: Average proportion of participants who visibly hesitated across each mistake level

Looking at the hesitation proportions first, we can see that the averages are slightly different between each mistake level. Surprisingly, the moderate mistake level had the fewest

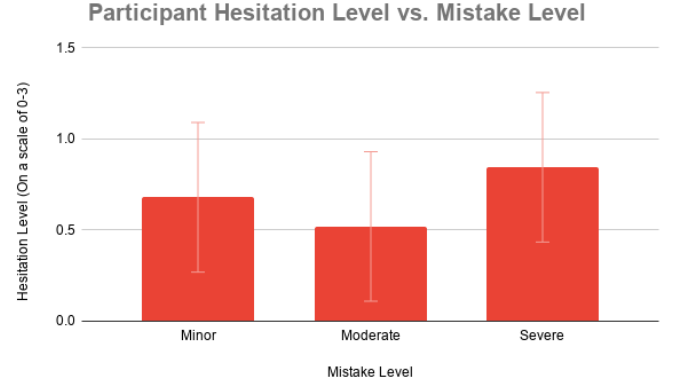


Fig. 5: Average hesitation level across three levels of mistakes

hesitations. The minor mistake level, which we predicted would have the lowest hesitation level, was second, then, as predicted, the severe had the highest hesitation proportion, with a proportion of 0.44. However, it is important to note through the error bars that the overall range of the data for each section is similar. This is similar for the hesitation levels, as they have slightly different means, but the overall ranges of data for each mistake level are similar.

This is further emphasized in the ANOVA tests for binary and level hesitation, comparing the means of each severity category. In both tests, the ANOVA test was not statistically significant, with a p-value of 0.1730 and 0.2306 for binary hesitation and hesitation level, respectively. Thus the tests failing to reject the null hypothesis that there is a statistical difference between any of the means of the three severity categories.

Along with the ANOVA results, we also computed pairwise Cohen's d effect sizes to assess the practical significance of these observed differences. The comparisons of each can be seen below:

The minor-vs-moderate comparison is close to the conventional small-to-medium threshold for both proportion and level of hesitation. The moderate vs. severe comparison yielded the largest effects of any pair, implying that the step from minor to moderate has less impact on hesitation behavior than moderate and severe or minor and severe.

Nevertheless, the trends in hesitation still show, implying that the behavior may exist across severity levels, but the current sample was insufficient to reliably represent this.

V. DISCUSSION

The goal of this study was to examine how varying levels of robot fault severity influence human trust in an HRI context. While prior work has established that robot errors negatively affect trust, our results extend this by exploring how the *degree* of those errors impacts both subjective perception and behavioral outcomes.

Cohen's d Value for Pairs	
Minor vs. Moderate	
Hesitation	Hesitation Level
0.5423261445	0.4352857501
Minor vs. Severe	
Hesitation	Hesitation Level
-0.3806246283	-0.3674029147
Moderate vs. Severe	
Hesitation	Hesitation Level
-1.08170489	-0.7593514298

Fig. 6: Pairwise Cohen's d effect sizes for binary hesitation and level across each different severity of mistakes.

A. Effect of Fault Severity on Trust

Our findings partially support the hypothesis that more severe robot errors lead to greater decreases in trust. Participants exposed to higher levels of fault severity consistently reported lower trust, reduced perceived reliability, and decreased willingness to follow the robot in future interactions. This aligns with prior findings that robot performance is a primary driver of perceived trustworthiness, as also shown in the original study where faulty behavior significantly reduced subjective trust ratings [1].

However, similar to the original study, this decrease in *subjective* trust did not always translate directly into changes in *behavior*. Even when participants recognized severe faults, many still complied with the robot's instructions, particularly in lower-stakes or reversible tasks. This reinforces the idea that trust in HRI is multidimensional: users may consciously distrust a robot while still complying with its requests.

B. Discrepancy Between Subjective and Behavioral Trust

One of the most important findings of this study is the clear discrepancy between subjective and objective measures of trust. While participants in the severe fault condition expressed significantly lower trust, hesitation times, and compliance behavior did not decrease proportionally.

This mirrors the findings of the original study, which reported that faulty robot behavior significantly impacted subjective evaluations but did not strongly influence compliance rates [1]. Together, these results suggest that behavioral trust is influenced by additional factors beyond perceived reliability, such as social pressure, task framing, or assumptions about the robot's authority.

As shown in Fig. 4, participants frequently looked around the room or toward the camera for guidance when the robot exhibited faults, suggesting uncertainty and a search for external confirmation. This led us to leave the room during trials so as not to accidentally influence the participants' actions in any way.

C. Role of Task Context

Task type emerged as a critical moderating factor. Participants were more hesitant or resistant when tasks were:

- Irreversible (e.g., pouring orange juice into a plant)
- Ethically questionable (e.g., privacy-related requests)

This supports prior findings that the nature of the request, particularly its perceived consequences, plays a larger role in compliance than robot performance alone [1]. In our study, even participants exposed to minor faults hesitated on high-stakes tasks, suggesting that trust is not solely dependent on robot reliability but also on human judgment of the situation.

D. Implications for HRI Design

These findings have several implications for the design of social robots:

- Error severity matters: Designers should minimize not just the frequency of errors, but also their perceived severity and visibility.
- Trust is not binary: Users may continue to comply with robots even when trust is degraded, which raises concerns about over-reliance.
- Context-aware interaction: Robots should adapt behavior based on task sensitivity, especially for high-stakes or ethically ambiguous requests.

Rather than aiming for perfect reliability, future systems may benefit from transparency mechanisms that help users appropriately calibrate trust.

E. Limitations

This study has several limitations. First, the sample size was relatively small and drawn from a population somewhat familiar with robotics, which may limit generalization. Second, the use of a Wizard-of-Oz setup may have reduced the perceived autonomy of the robot. Third, our measurement of objective trust relied primarily on hesitation timing rather than binary compliance, which may not fully capture behavioral nuances.

Additionally, the Misty II robot differs from the Sunflower robot used in the original study in size and embodiment, which may influence anthropomorphism and trust perception.

F. Future Work

Future research should explore:

- Larger and more diverse participant populations
- Longitudinal interactions to study trust over time
- More granular behavioral measures (e.g., eye tracking, physiological responses)
- The role of explanation or apology strategies in mitigating severe faults

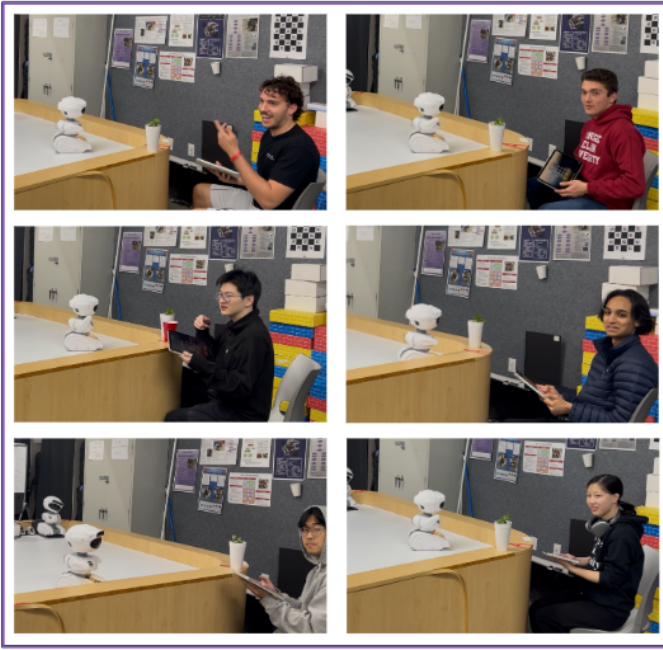


Fig. 7: Participants occasionally glance towards the camera or around the room to attempt to look for help when the robot errs

Further investigation into how users reconcile conflicting subjective and behavioral trust signals would also provide valuable insight into trust calibration in HRI.

VI. CONCLUSION

This study investigated how varying levels of robot fault severity influence human trust in an HRI setting. By extending prior work that primarily compared correct versus faulty behavior, we introduced multiple levels of error severity to better understand how trust changes along a spectrum rather than as a binary outcome.

Our findings show that increased fault severity leads to a clear decrease in *subjective* trust, with participants reporting lower perceived reliability and reduced confidence in the robot as errors became more pronounced. However, this decrease in perceived trust did not consistently translate into reduced *behavioral* compliance. Participants often continued to follow the robot's instructions even when they recognized significant faults, particularly in tasks that were low-risk or easily reversible.

The disconnect between subjective and objective trust showed that trust in HRI is a combination of perception, context, and human decision-making rather than a single construct. In particular, task type plays a critical role in moderating behavior, as participants were more likely to resist requests that were irreversible or ethically questionable, regardless of the robot's performance level.

These results suggest that designing trustworthy robots requires more than simply improving technical reliability. Systems must also support appropriate trust calibration by

accounting for error severity, task context, and user perception. Future work should continue to explore how these factors interact over longer-term interactions and in more diverse populations, with the goal of developing robots that users neither over-trust nor under-trust, but rely on appropriately.

While more severe robot faults reduced perceived trust, they did not linearly reduce compliance, demonstrating the complexity of trust in human-robot interaction and the importance of designing systems that align user perception with behavior.

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